

Code 1:

```
# Diffie-Hellman Key Exchange
# Educational implementation
```

```
# Public parameters
```

```
p = 23  # Prime number
```

```
g = 5   # Primitive root / generator
```

```
# Private keys (kept secret)
```

```
hari_private = 6
```

```
dhairya_private = 15
```

```
# Public keys
```

```
hari_public = pow(g, hari_private, p)
```

```
dhairya_public = pow(g, dhairya_private, p)
```

```
print("Hari's public key:", hari_public)
```

```
print("Dhairya's public key:", dhairya_public)
```

```
# Each party calculates the shared secret
```

```
hari_shared_secret = pow(dhairya_public, hari_private, p)
```

```
dhairya_shared_secret = pow(hari_public, dhairya_private, p)
```

```
print("Hari's shared secret:", hari_shared_secret)
```

```
print("Dhairya's shared secret:", dhairya_shared_secret)
```

```
# Verify that both calculated the same secret
```

```
if hari_shared_secret == dhairya_shared_secret:
```

```
    print("Key exchange successful!")
```

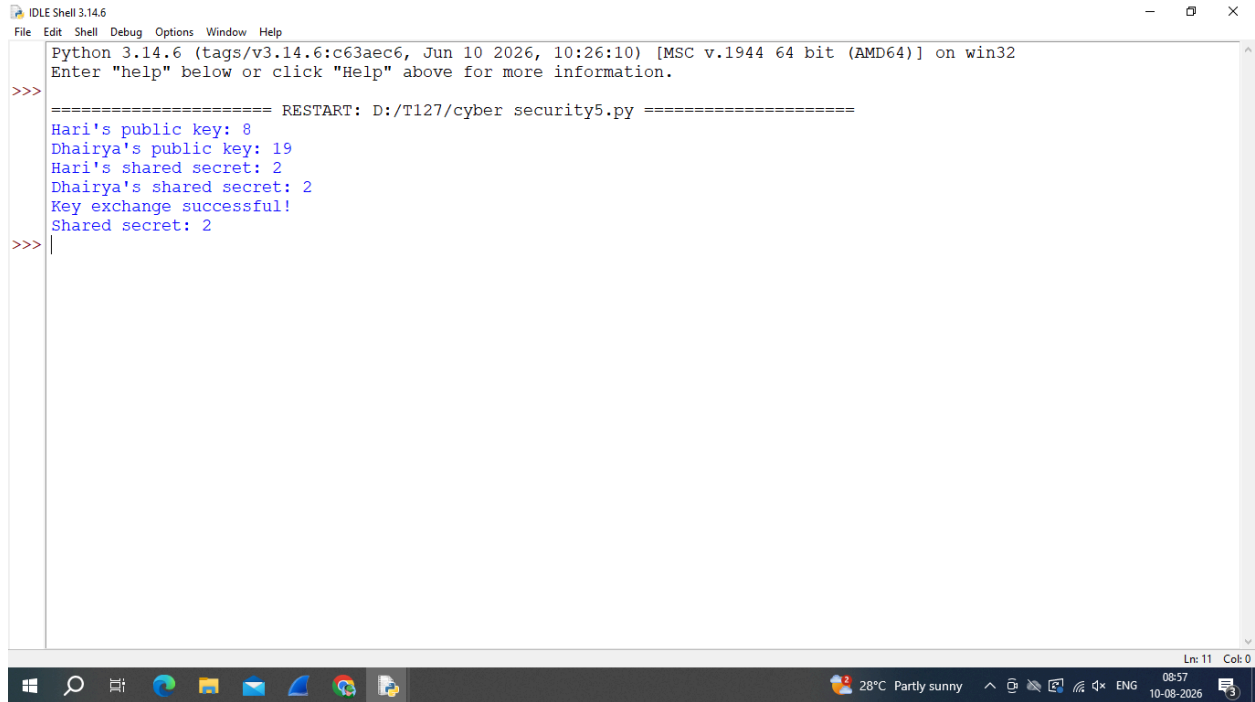
```
    print("Shared secret:", hari_shared_secret)
```

```
else:
```

```
    print("Key exchange failed!")
```

Output1:

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The screenshot shows a Python IDLE Shell window titled 'IDLE Shell 3.14.6'. The menu bar includes File, Edit, Shell, Debug, Options, Window, and Help. The shell displays the following text:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/T127/cyber security5.py =====
Hari's public key: 8
Dhairya's public key: 19
Hari's shared secret: 2
Dhairya's shared secret: 2
Key exchange successful!
Shared secret: 2
>>>
```

The taskbar at the bottom shows the Windows Start button, search icon, and several application icons. The system tray on the right indicates a temperature of 28°C, 'Partly sunny' weather, and the date '10-08-2026'.

Code2:

```
import tkinter as tk
from tkinter import messagebox
```

```
def perform_diffie_hellman():
    try:
        # Get values from GUI
        p = int(p_entry.get())
        g = int(g_entry.get())
        alice_private = int(alice_entry.get())
        bob_private = int(bob_entry.get())

        # Basic validation
        if p <= 1:
            raise ValueError("p must be greater than 1.")

        if g <= 0:
            raise ValueError("g must be positive.")

        if alice_private <= 0 or bob_private <= 0:
            raise ValueError("Private keys must be positive.")

        # Calculate public keys
        alice_public = pow(g, alice_private, p)
```

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    bob_public = pow(g, bob_private, p)

    # Calculate shared secret
    alice_shared = pow(bob_public, alice_private, p)
    bob_shared = pow(alice_public, bob_private, p)

    # Display results
    alice_public_result.config(text=str(alice_public))
    bob_public_result.config(text=str(bob_public))
    alice_shared_result.config(text=str(alice_shared))
    bob_shared_result.config(text=str(bob_shared))

    if alice_shared == bob_shared:
        status_label.config(
            text="✓ Key Exchange Successful!",
            fg="green"
        )
    else:
        status_label.config(
            text="✗ Key Exchange Failed!",
            fg="red"
        )

except ValueError as e:
    messagebox.showerror("Invalid Input", str(e))

def clear_fields():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    alice_entry.delete(0, tk.END)
    bob_entry.delete(0, tk.END)

    alice_public_result.config(text="-")
    bob_public_result.config(text="-")
    alice_shared_result.config(text="-")
    bob_shared_result.config(text="-")

    status_label.config(text="")

# ----- GUI -----

```

```
root.title("Diffie-Hellman Key Exchange")
root.geometry("600x600")
root.resizable(False, False)
root.configure(bg="#f2f2f2")

# Title
title = tk.Label(
    root,
    text="Diffie-Hellman Key Exchange",
    font=("Arial", 22, "bold"),
    bg="#f2f2f2",
    fg="#1f3c88"
)
title.pack(pady=20)

subtitle = tk.Label(
    root,
    text="Secure key exchange demonstration",
    font=("Arial", 11),
    bg="#f2f2f2",
    fg="#555555"
)
subtitle.pack()

# Input frame
input_frame = tk.Frame(root, bg="white", padx=20, pady=20)
input_frame.pack(pady=20)

# Public parameters
tk.Label(
    input_frame,
    text="Public Parameters",
    font=("Arial", 14, "bold"),
    bg="white"
).grid(row=0, column=0, columnspan=2, pady=10)

tk.Label(
    input_frame,
    text="Prime number (p):",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, sticky="w", pady=7)

p_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
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p_entry.grid(row=1, column=1, pady=7)

tk.Label(
    input_frame,
    text="Generator (g):",
    font=("Arial", 11),
    bg="white"
).grid(row=2, column=0, sticky="w", pady=7)

g_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
g_entry.grid(row=2, column=1, pady=7)

# Private keys
tk.Label(
    input_frame,
    text="Private Keys",
    font=("Arial", 14, "bold"),
    bg="white"
).grid(row=3, column=0, columnspan=2, pady=(15, 10))

tk.Label(
    input_frame,
    text="Alice's private key:",
    font=("Arial", 11),
    bg="white"
).grid(row=4, column=0, sticky="w", pady=7)

alice_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
alice_entry.grid(row=4, column=1, pady=7)

tk.Label(
    input_frame,
    text="Bob's private key:",
    font=("Arial", 11),
    bg="white"
).grid(row=5, column=0, sticky="w", pady=7)

bob_entry = tk.Entry(input_frame, width=25, font=("Arial", 11))
bob_entry.grid(row=5, column=1, pady=7)

# Buttons
button_frame = tk.Frame(root, bg="#f2f2f2")
button_frame.pack(pady=5)

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calculate_button = tk.Button(
    button_frame,
    text="Calculate Shared Key",
    command=perform_diffie_hellman,
    bg="#1f6feb",
    fg="white",
    font=("Arial", 11, "bold"),
    padx=15,
    pady=8
)
calculate_button.grid(row=0, column=0, padx=10)

clear_button = tk.Button(
    button_frame,
    text="Clear",
    command=clear_fields,
    bg="#555555",
    fg="white",
    font=("Arial", 11, "bold"),
    padx=25,
    pady=8
)
clear_button.grid(row=0, column=1, padx=10)

# Results
result_frame = tk.Frame(root, bg="white", padx=25, pady=15)
result_frame.pack(pady=15)

tk.Label(
    result_frame,
    text="Results",
    font=("Arial", 14, "bold"),
    bg="white"
).grid(row=0, column=0, columnspan=2, pady=5)

tk.Label(
    result_frame,
    text="Alice's Public Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, sticky="w", pady=5)

alice_public_result = tk.Label(
    result_frame,
    text="Alice's Public Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=1, sticky="w", pady=5)

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text="-",
font=("Arial", 11, "bold"),
bg="white",
fg="#1f3c88"
)
alice_public_result.grid(row=1, column=1, sticky="w", padx=20)
```

```
tk.Label(
    result_frame,
    text="Bob's Public Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=2, column=0, sticky="w", pady=5)
```

```
bob_public_result = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11, "bold"),
    bg="white",
    fg="#1f3c88"
)
bob_public_result.grid(row=2, column=1, sticky="w", padx=20)
```

```
tk.Label(
    result_frame,
    text="Alice's Shared Secret:",
    font=("Arial", 11),
    bg="white"
).grid(row=3, column=0, sticky="w", pady=5)
```

```
alice_shared_result = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11, "bold"),
    bg="white",
    fg="#8a2be2"
)
alice_shared_result.grid(row=3, column=1, sticky="w", padx=20)
```

```
tk.Label(
    result_frame,
    text="Bob's Shared Secret:",
    font=("Arial", 11),
    bg="white"
```

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).grid(row=4, column=0, sticky="w", pady=5)

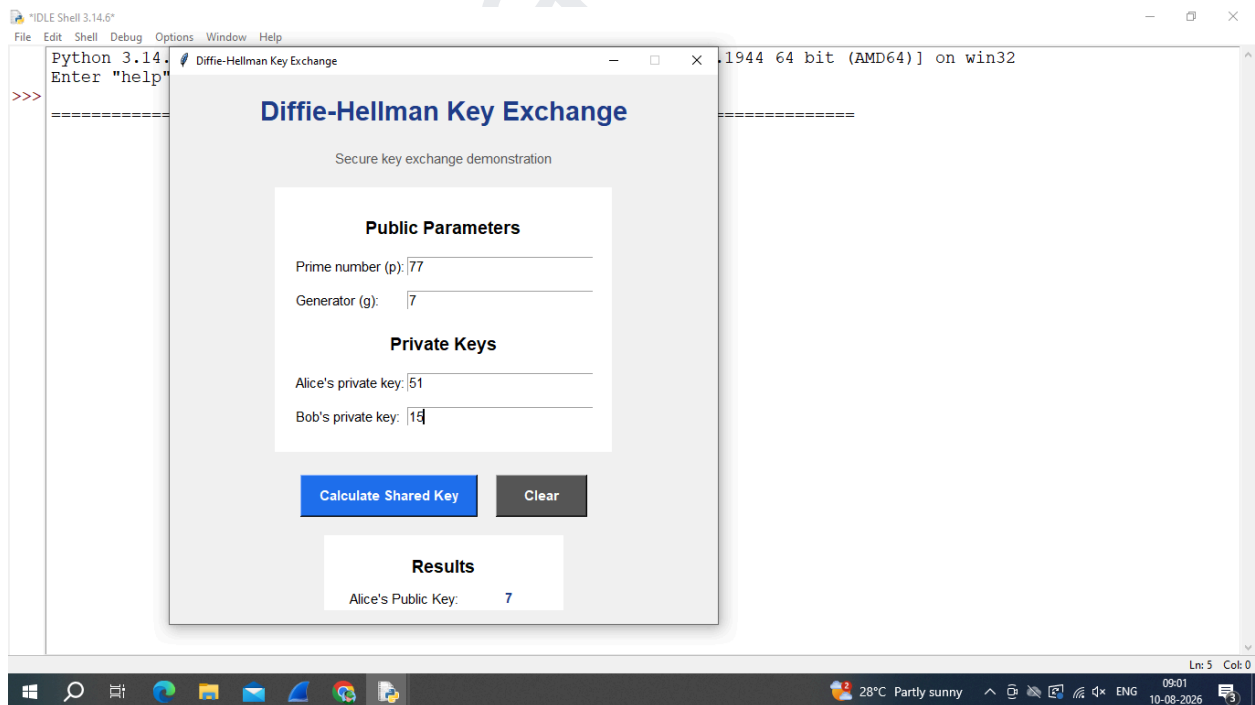
bob_shared_result = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11, "bold"),
    bg="white",
    fg="#8a2be2"
)
bob_shared_result.grid(row=4, column=1, sticky="w", padx=20)

# Status
status_label = tk.Label(
    root,
    text="",
    font=("Arial", 13, "bold"),
    bg="#f2f2f2"
)
status_label.pack(pady=10)

root.mainloop()

```

Output2:



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